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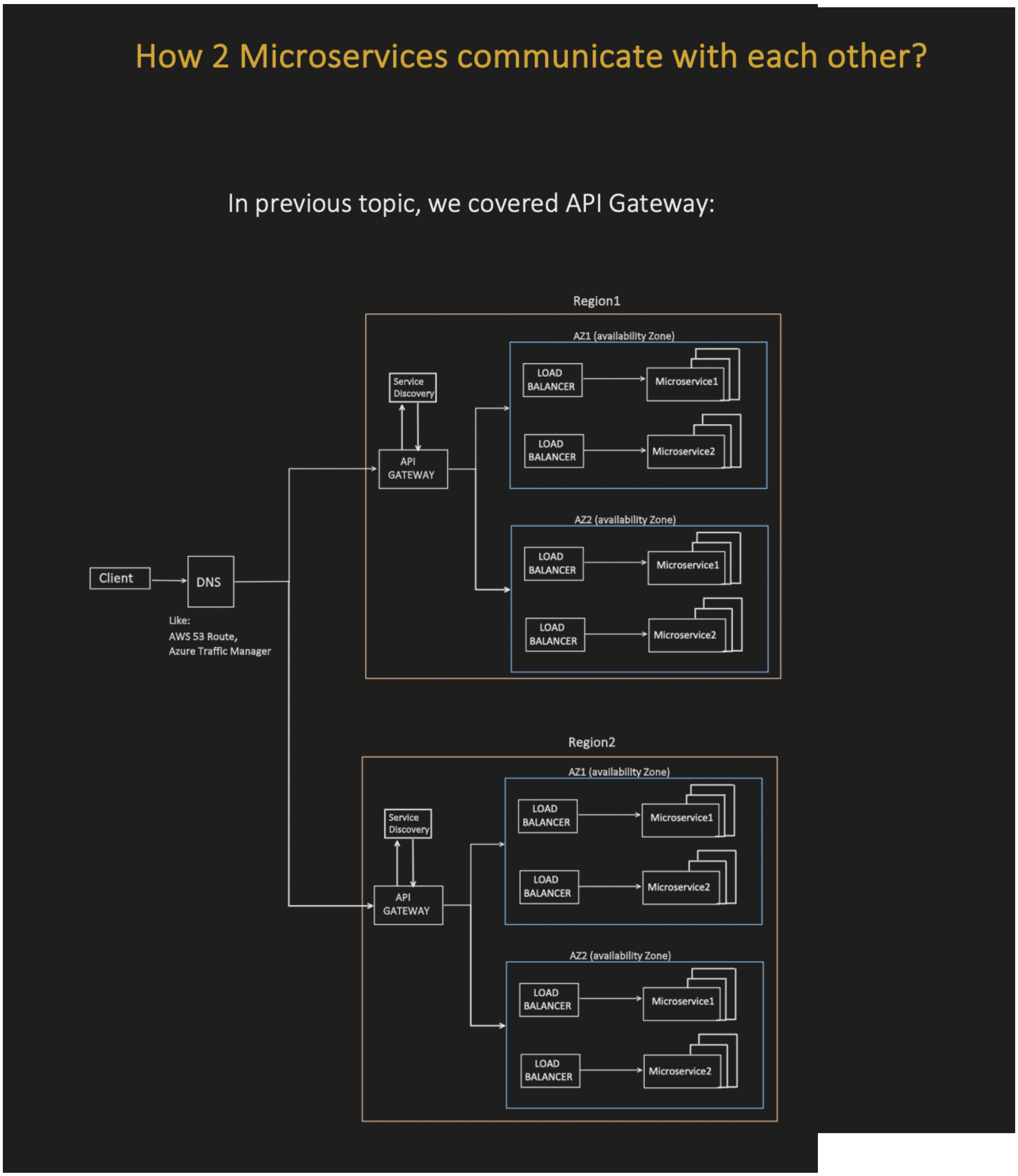
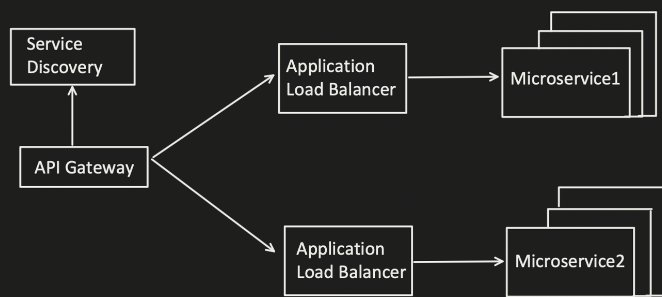
HLD: Service Mesh



In Simple, above diagram can be represented as:



Now, follow up question which comes is: How 2 different Microservices communicate with each other?

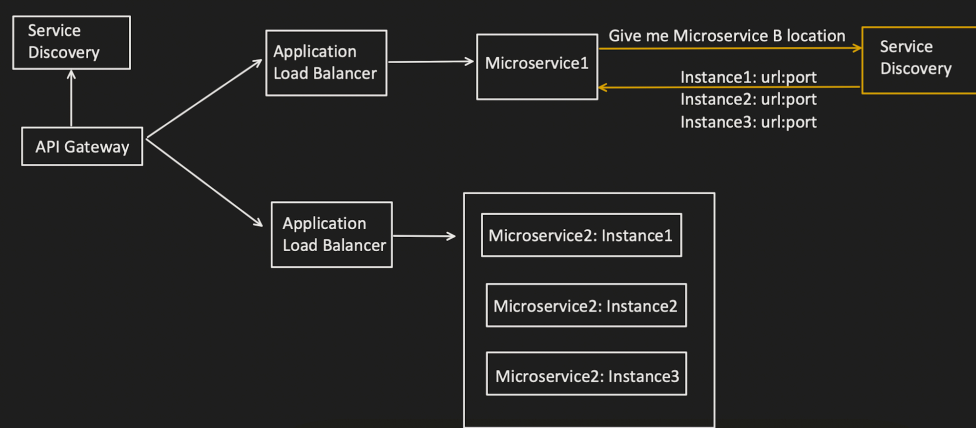
Lets understand the need, with example: Microservice A want to communicate with Microservice B

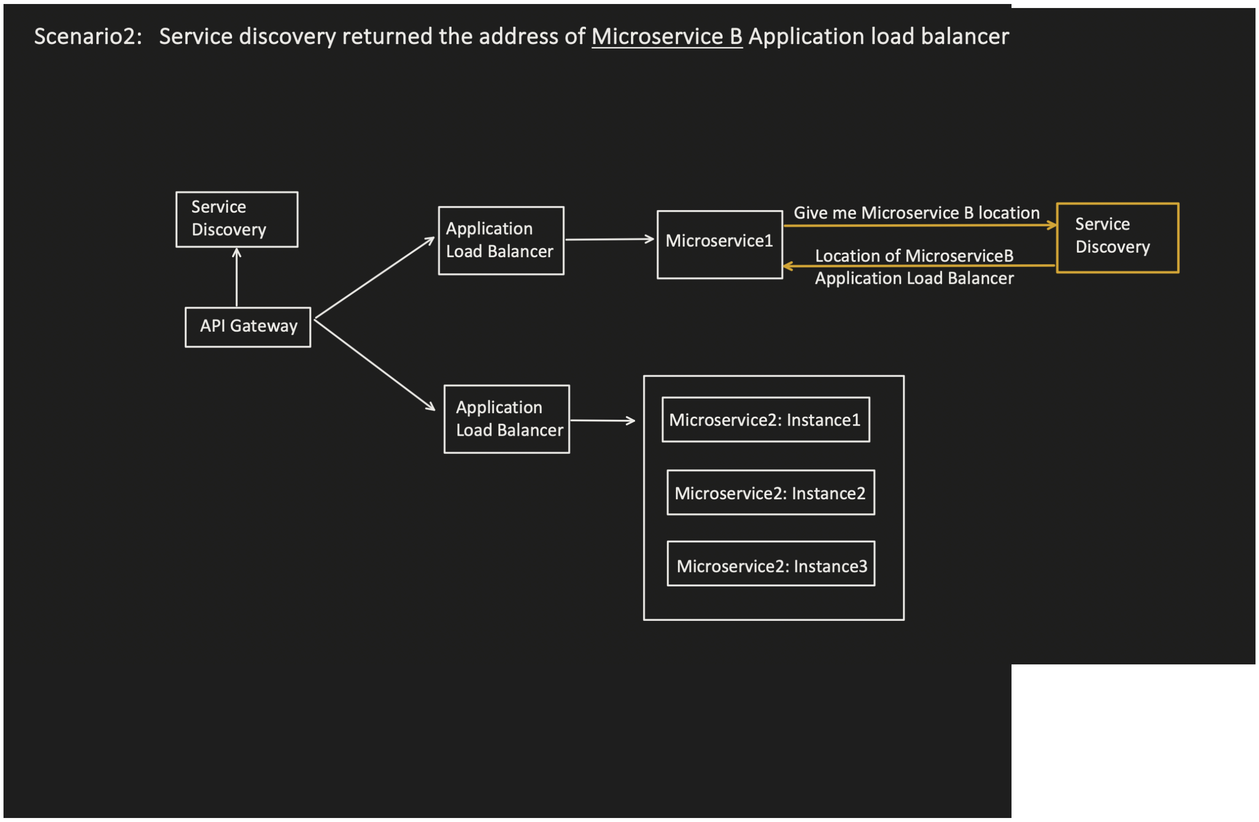
1. Service Discovery Capability :

- Microservice A wants to communicate with Microservice B. Then it need its location (IP address and port no).

- Service Discovery functionality is required so that Microservice A can get the location of Microservice B different instances.

Scenario1: Service Discovery returns all the Microservice B instances location.



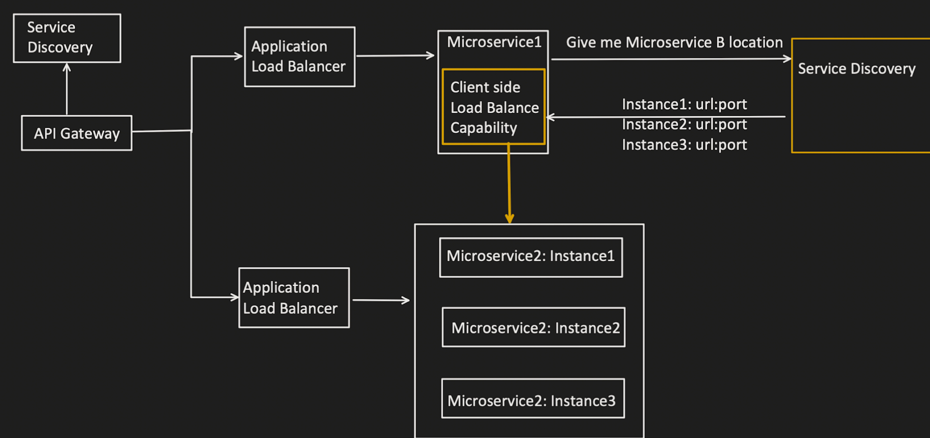

Scenario2: Service discovery returned the address of Microservice B Application load balancer
Service Discovery
API Gateway
Application Load Balancer
Microservice1
Give me Microservice B location
Location of MicroserviceB Application Load Balancer
Service Discovery
Application Load Balancer
Microservice2: Instance1
Microservice2: Instance2
Microservice2: Instance3

2. Load Balancing Capability:

Scenario1:

Microservice A got the Microservice B instances location and now want Load Balancing capability to equally distribute the traffic to Microservice B.

application.properties
microserviceB.url={instance-url}:{port-no}

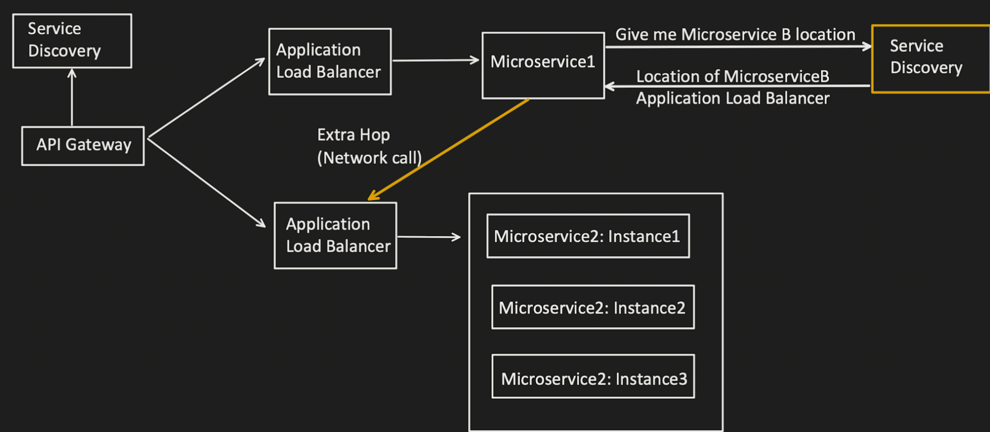


Scenario2:

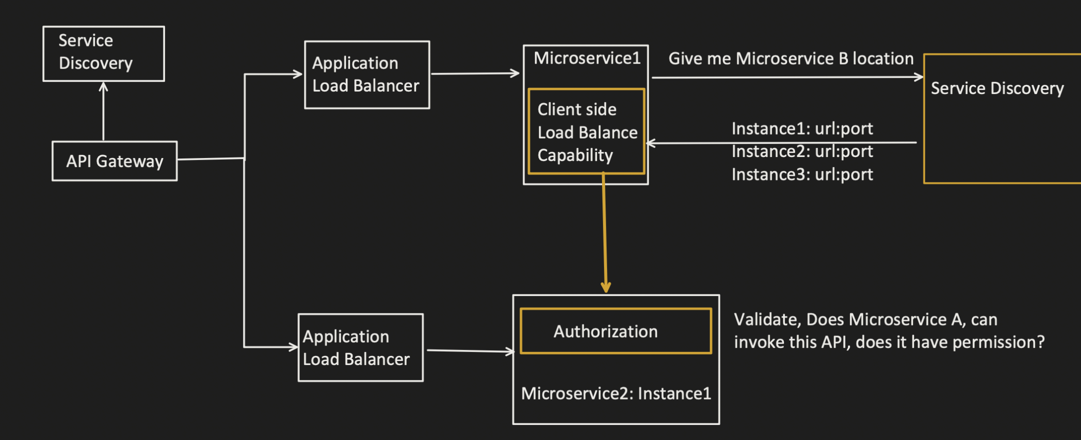
- Microservice A got the Microservice B Application load balancer address and now need to invoke that.
- Extra Network call is required, which might cause latency issue.

application.properties

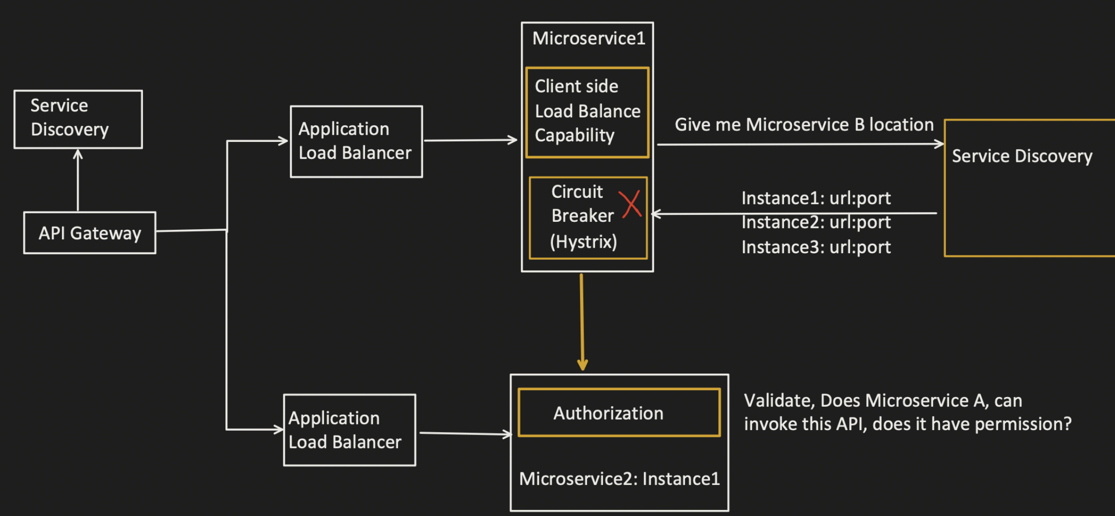
microserviceB.url={application-load-balancer-url}:{port-no}



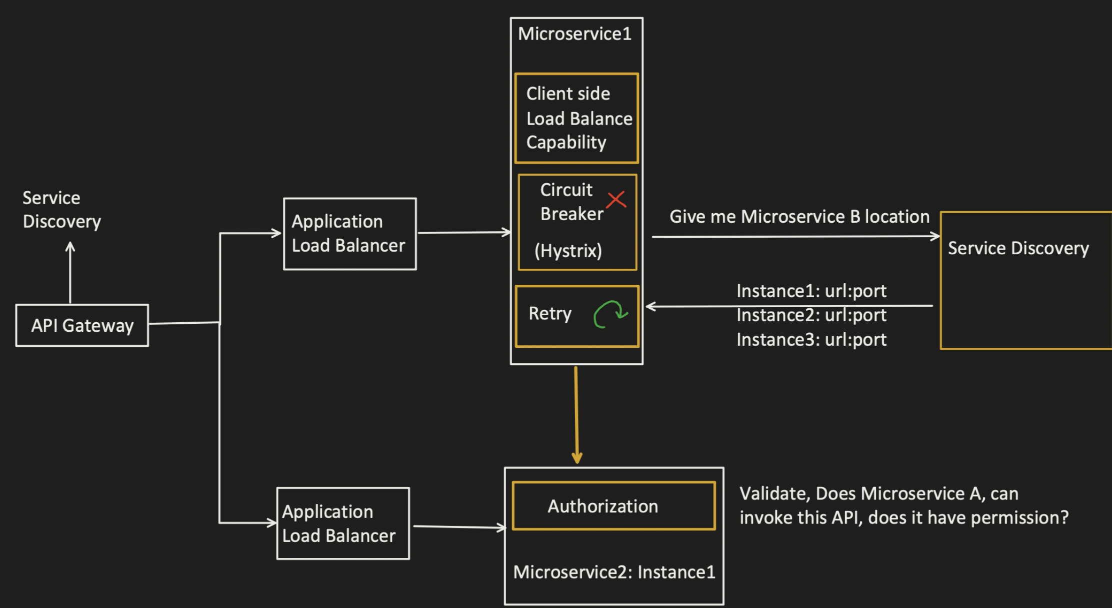
3.Authorization:



4. Circuit Breaker Capability:



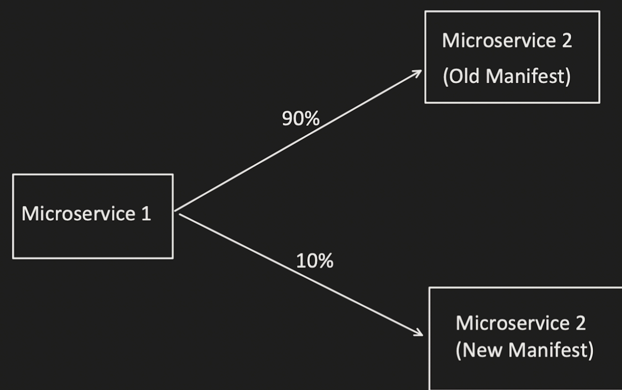
5. Re-try Capability:



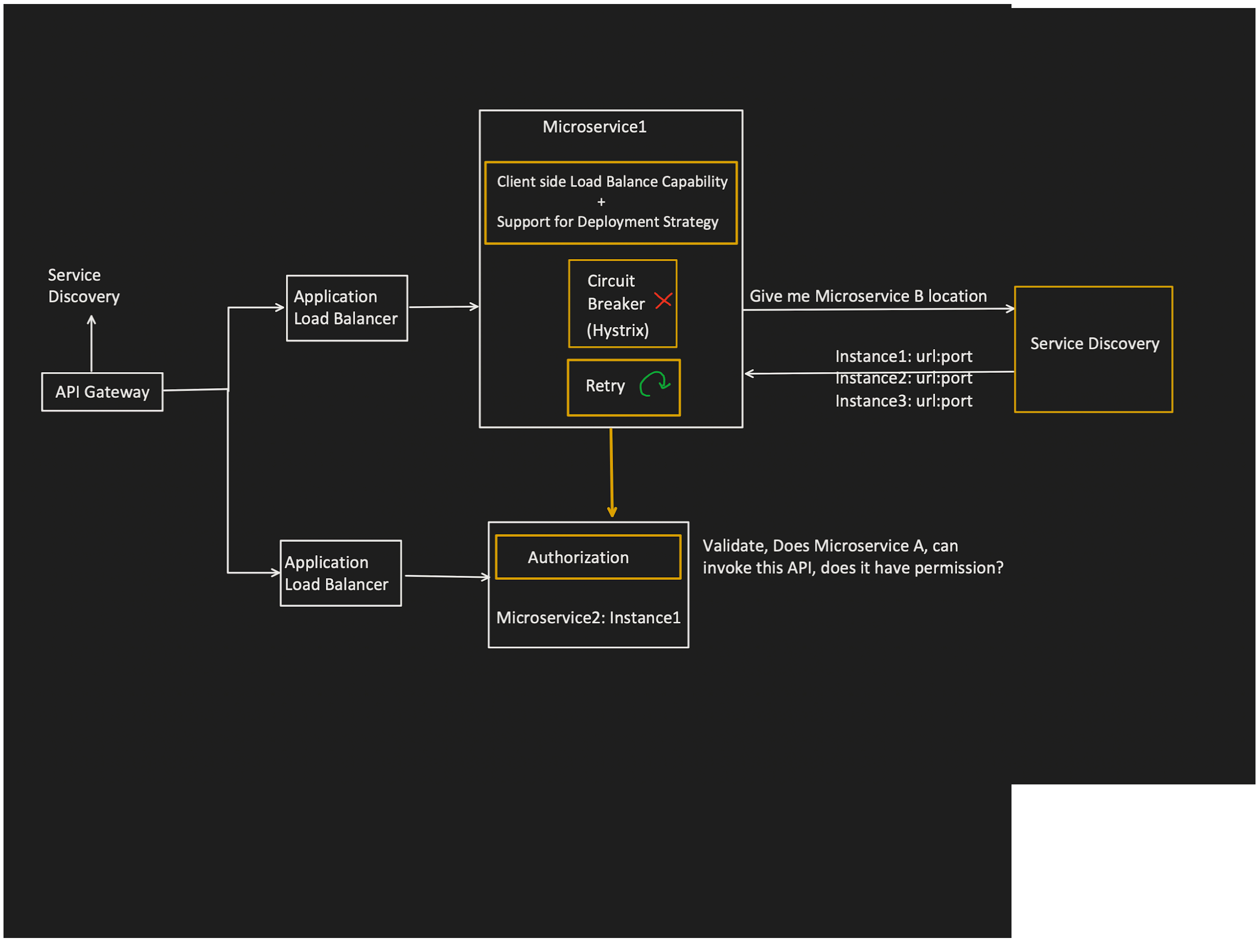
6. Deployment Strategy:

There are different deployment strategy like Canary, Red-Black, Blue-Green etc.

For example: In Canary, I want 90% traffic on Old and 10% on new initially, and gradually increase the number on new manifest.

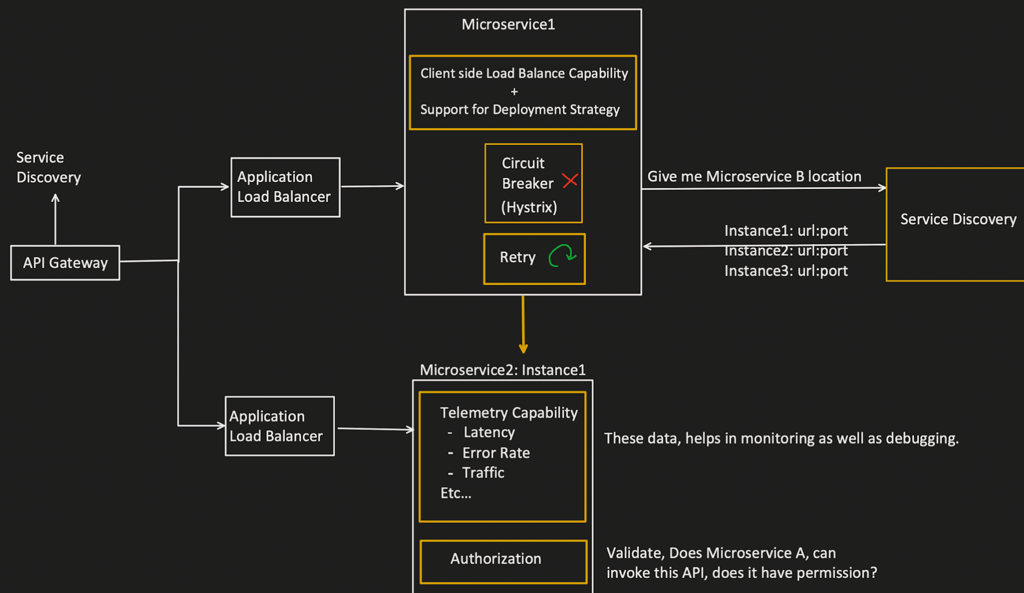


Generally Load balancer helps to control this, means "Client side load balancer also has to handle this different deployment strategy"


Service
Discovery
API Gateway
Application
Load Balancer
Application
Load Balancer
Microservice1
Client side Load Balance Capability
+
Support for Deployment Strategy
Circuit
Breaker
(Hystrix)
Retry
Give me Microservice B location
Service Discovery
Instance1: url:port
Instance2: url:port
Instance3: url:port
Authorization
Microservice2: Instance1
Validate, Does Microservice A, can
invoke this API, does it have permission?

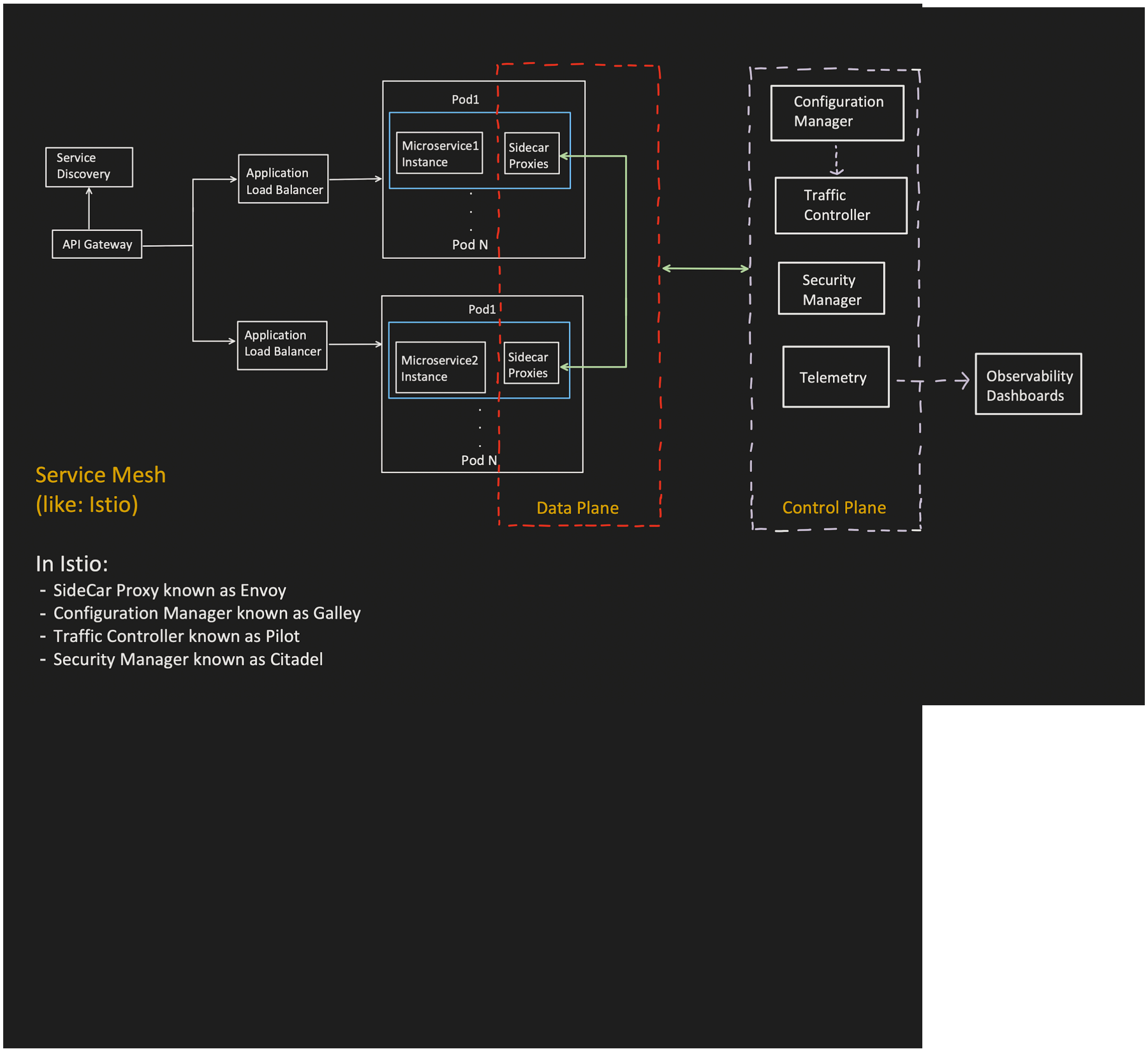
7. Telemetry Capability:

It means, we need to collect data which will help in analysis and monitoring of a component.



Any better solution?

Answer is : SERVICE MESH



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